1

Problem Description:

How to split a string into a number of substrings ?

Solution:

Following example shows how to change the host name to its specific IP address with the help of InetAddress.getByName() method of net.InetAddress class.

```
import java.net.InetAddress;
import java.net.UnknownHostException;

public class GetIP {
   public static void main(String[] args) {
      InetAddress address = null;
      try {
         address = InetAddress.getByName
         ("www.javatutorial.com");
      }
           catch (UnknownHostException e) {
         System.exit(2);
      }
      System.out.println(address.getHostName() + "="
      + address.getHostAddress());
      System.exit(0);
   }
}
```

Result:

The above code sample will produce the following result.

```
http://www.javatutorial.com = 123.14.2.35
```

2

Problem Description:

How to get connected with web server?

Solution:

Following example demonstrates how to get connected with web server by using sock.getInetAddress() method of net.Socket class.

```
import java.net.InetAddress;
import java.net.Socket;

public class WebPing {
   public static void main(String[] args) {
      try {
         InetAddress addr;
         Socket sock = new Socket("www.javatutorial.com", 80);
         addr = sock.getInetAddress();
         System.out.println("Connected to " + addr);
         sock.close();
      } catch (java.io.IOException e) {
         System.out.println("Can't connect to " + args[0]);
         System.out.println(e);
      }
   }
}
```

Result:

The above code sample will produce the following result.

```
Connected to http://www,javatutorial.com/102.32.56.14
```

3

Problem Description:

How to check a file is modified at a server or not?

Solution:

Following example shows How to check a file is modified at a server or not.

```
import java.net.URL;
import java.net.URLConnection;

public class Main {
   public static void main(String[] argv)
   throws Exception {
      URL u = new URL("http://127.0.0.1/java.bmp");
      URLConnection uc = u.openConnection();
      uc.setUseCaches(false);
      long timestamp = uc.getLastModified();
      System.out.println("The last modification time
      of java.bmp is :"+timestamp);
   }
}
```

Result:

The above code sample will produce the following result.

```
The last modification time of java.bmp is :24 May 2009 12:14:50
```

4

Problem Description:

How to create a multithreaded server?

Solution:

Following example demonstrates how to create a multithreaded server by using ssock.accept() method of Socket class and MultiThreadServer(socketname) method of ServerSocket class.

```
import java.io.IOException;
import java.io.PrintStream;
import java.net.ServerSocket;
import java.net.Socket;

public class MultiThreadServer implements Runnable {
   Socket csocket;
   MultiThreadServer(Socket csocket) {
      this.csocket = csocket;
   }

   public static void main(String args[])
   throws Exception {
      ServerSocket ssock = new ServerSocket(1234);
      System.out.println("Listening");
      while (true) {
         Socket sock = ssock.accept();
         System.out.println("Connected");
         new Thread(new MultiThreadServer(sock)).start();
      }
   }
   public void run() {
      try {
         PrintStream pstream = new PrintStream
         (csocket.getOutputStream());
         for (int i = 100; i >= 0; i--) {
            pstream.println(i +
            " bottles of beer on the wall");
         }
         pstream.close();
         csocket.close();
      }
      catch (IOException e) {
         System.out.println(e);
      }
   }
}
```

Result:

The above code sample will produce the following result.

```
Listening
Connected
```

5

Problem Description:

How to get the file size from the server?

Solution:

Following example demonstrates How to get the file size from the server.

```
import java.net.URL;
import java.net.URLConnection;

public class Main {
   public static void main(String[] argv)
   throws Exception {
      int size;
      URL url = new URL("http://www.server.com");
      URLConnection conn = url.openConnection();
      size = conn.getContentLength();
      if (size < 0)
      System.out.println("Could not determine file size.");
      else
      System.out.println("The size of file is = "
      +size + "bytes");
      conn.getInputStream().close();
   }
}
```

Result:

The above code sample will produce the following result.

```
The size of file is = 16384 bytes
```

6

Problem Description:

How to make a socket displaying message to a single client?

Solution:

Following example demonstrates how to make a socket displaying message to a single client with the help of ssock.accept() method of Socket class.

```
import java.io.PrintStream;
import java.net.ServerSocket;
import java.net.Socket;

public class BeerServer {
   public static void main(String args[])
   throws Exception {
      ServerSocket ssock = new ServerSocket(1234);
      System.out.println("Listening");
      Socket sock = ssock.accept();
      ssock.close();
      PrintStream ps = new PrintStream
      (sock.getOutputStream());
      for (int i = 10; i >= 0; i--) {
         ps.println(i +
         " from Java Source and Support.");
      }
      ps.close();
      sock.close();
   }
}
```

Result:

The above code sample will produce the following result.

```
Listening
10 from Java Source and Support
9 from Java Source and Support
8 from Java Source and Support
7 from Java Source and Support
6 from Java Source and Support
5 from Java Source and Support
4 from Java Source and Support
3 from Java Source and Support
2 from Java Source and Support
1 from Java Source and Support
0 from Java Source and Support
```

7

Problem Description:

How to make a server to allow the connection to the socket 6123?

Solution:

Following example shows how to make a server to allow the connection to the socket 6123 by using server.accept() method of ServerSocket class and sock.getInetAddress() method of Socket class.

```
import java.io.IOException;
import java.net.InetAddress;
import java.net.ServerSocket;
import java.net.Socket;

public class SocketDemo {
   public static void main(String[] args) {
      try {
         ServerSocket server = new ServerSocket(6123);
         while (true) {
            System.out.println("Listening");
            Socket sock = server.accept();
            InetAddress addr = sock.getInetAddress();
            System.out.println("Connection made to "
            + addr.getHostName()
            + " (" + addr.getHostAddress() + ")");
            pause(5000);
            sock.close();
         }
      }
           catch (IOException e) {
         System.out.println("Exception detected: " + e);
      }
   }
   private static void pause(int ms) {
      try {
         Thread.sleep(ms);
      }
      catch (InterruptedException e) {
      }
   }
}
```

Result:

The above code sample will produce the following result.

```
Listening
Terminate batch job (Y/N)? n
Connection made to 112.63.21.45
```

8

Problem Description:

How to get the parts of an URL?

Solution:

Following example shows how to get the parts of an URL with the help of url.getProtocol() ,url.getFile() method etc. of net.URL class.

```
import java.net.URL;

public class Main {
   public static void main(String[] args)
   throws Exception {
      URL url = new URL(args[0]);
      System.out.println("URL is " + url.toString());
      System.out.println("protocol is "
      + url.getProtocol());
      System.out.println("file name is " + url.getFile());
      System.out.println("host is " + url.getHost());
      System.out.println("path is " + url.getPath());
      System.out.println("port is " + url.getPort());
      System.out.println("default port is "
      + url.getDefaultPort());
   }
}
```

Result:

The above code sample will produce the following result.

```
URL is http://www.server.com
protocol is TCP/IP
file name is java_program.txt
host is 122.45.2.36
path is
port is 2
default port is 1
```

9

Problem Description:

How to get the date of URL connection?

Solution:

Following example demonstrates how to get the date of URL connection by using httpCon.getDate() method of HttpURLConnection class.

```
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.Date;

public class Main{
   public static void main(String args[])
   throws Exception {
      URL url = new URL("http://www.google.com");
      HttpURLConnection httpCon =
      (HttpURLConnection) url.openConnection();
      long date = httpCon.getDate();
      if (date == 0)
      System.out.println("No date information.");
      else
      System.out.println("Date: " + new Date(date));
   }
}
```

Result:

The above code sample will produce the following result.

```
Date:05.26.2009
```

10

Problem Description:

How to read and download a webpage?

Solution:

Following example shows how to read and download a webpage URL() constructer of net.URL class.

```
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileWriter;
import java.io.InputStreamReader;
import java.net.URL;

public class Main {
   public static void main(String[] args)
   throws Exception {
      URL url = new URL("http://www.google.com");
      BufferedReader reader = new BufferedReader
      (new InputStreamReader(url.openStream()));
      BufferedWriter writer = new BufferedWriter
      (new FileWriter("data.html"));
      String line;
      while ((line = reader.readLine()) != null) {
         System.out.println(line);
         writer.write(line);
         writer.newLine();
      }
      reader.close();
      writer.close();
   }
}
```

Result:

The above code sample will produce the following result.

```
Welcome to Java Tutorial
        Here we have plenty of examples for you!

        Come and Explore Java!
```

11

Problem Description:

How to find hostname from IP Address?

Solution:

Following example shows how to change the host name to its specific IP address with the help of InetAddress.getByName() method of net.InetAddress class.

```
import java.net.InetAddress;

public class Main {
  public static void main(String[] argv) throws Exception {

    InetAddress addr = InetAddress.getByName("74.125.67.100");
    System.out.println("Host name is: "+addr.getHostName());
    System.out.println("Ip address is: "+ addr.getHostAddress());
  }
}
```

Result:

The above code sample will produce the following result.

```
http://www.javatutorial.com = 123.14.2.35
```

12

Problem Description:

How to determine IP Address & hostname of Local Computer?

Solution:

Following example shows how to find local IP Address & Hostname of the system using getLocalAddress() method of InetAddress class.

```
import java.net.InetAddress;

public class Main {
   public static void main(String[] args)
   throws Exception {
      InetAddress addr = InetAddress.getLocalHost();
      System.out.println("Local HostAddress:
      "+addr.getHostAddress());
      String hostname = addr.getHostName();
      System.out.println("Local host name: "+hostname);
   }
}
```

Result:

The above code sample will produce the following result.

```
Local HostAddress: 192.168.1.4
Local host name: harsh
```

13

Problem Description:

How to check whether a port is being used or not?

Solution:

Following example shows how to check whether any port is being used as a server or not by creating a socket object.

```
import java.net.*;
import java.io.*;

public class Main {
   public static void main(String[] args) {
      Socket Skt;
      String host = "localhost";
      if (args.length gt; 0) {
         host = args[0];
      }
      for (int i = 0; i < 1024; i++) {
         try {
            System.out.println("Looking for "+ i);
            Skt = new Socket(host, i);
            System.out.println("There is a server on port "
            + i + " of " + host);
         }
         catch (UnknownHostException e) {
            System.out.println("Exception occured"+ e);
            break;
         }
         catch (IOException e) {
         }
      }
   }
}
```

Result:

The above code sample will produce the following result.

```
Looking for 0
Looking for 1
Looking for 2
Looking for 3
Looking for 4. . .
```

14

Problem Description:

How to find proxy settings of a System?

Solution:

Following example shows how to find proxy settings & create a proxy connection on a system using put method of systemSetting & getResponse method of HttpURLConnection class.

```
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.Properties;
import java.net.InetSocketAddress;
import java.net.Proxy;
import java.net.ProxySelector;
import java.net.URI;

public class Main{
   public static void main(String s[])
   throws Exception{
      try {
         Properties systemSettings =
         System.getProperties();
         systemSettings.put("proxySet", "true");
         systemSettings.put("http.proxyHost",
         "proxy.mycompany1.local");
         systemSettings.put("http.proxyPort", "80");
         URL u = new URL("http://www.google.com");
         HttpURLConnection con = (HttpURLConnection)
         u.openConnection();
         System.out.println(con.getResponseCode() +
         " : " + con.getResponseMessage());
         System.out.println(con.getResponseCode() ==
         HttpURLConnection.HTTP_OK);
      }
      catch (Exception e) {
         e.printStackTrace();
         System.out.println(false);
      }
      System.setProperty("java.net.useSystemProxies",
      "true");
      Proxy proxy = (Proxy) ProxySelector.getDefault().
      select(new URI("http://www.yahoo.com/")).iterator().
      next();;
      System.out.println("proxy hostname : " + proxy.type());
      InetSocketAddress addr = (InetSocketAddress)
      proxy.address();
      if (addr == null) {
         System.out.println("No Proxy");
      }
      else {
         System.out.println("proxy hostname : "
         + addr.getHostName());
         System.out.println("proxy port : "
         + addr.getPort());
      }
   }
}
```

Result:

The above code sample will produce the following result.

```
200 : OK
true
proxy hostname : HTTP
proxy hostname : proxy.mycompany1.local
proxy port : 80
```

15

Problem Description:

How to create a socket at a specific port?

Solution:

Following example shows how to sing Socket constructor of Socket class.And also get Socket details using getLocalPort() getLocalAddress , getInetAddress() & getPort() methods.

```
import java.io.IOException;
import java.net.InetAddress;
import java.net.Socket;
import java.net.SocketException;
import java.net.UnknownHostException;

public class Main {
   public static void main(String[] args) {
      try {
         InetAddress addr =
         InetAddress.getByName("74.125.67.100");
         Socket theSocket = new Socket(addr, 80);
         System.out.println("Connected to "
         + theSocket.getInetAddress()
         + " on port " + theSocket.getPort() + " from port "
         + theSocket.getLocalPort() + " of "
         + theSocket.getLocalAddress());
      }
      catch (UnknownHostException e) {
         System.err.println("I can't find " + e  );
      }
      catch (SocketException e) {
         System.err.println("Could not connect to " +e );
      }
      catch (IOException e) {
         System.err.println(e);
      }
   }
}
```

Result:

The above code sample will produce the following result.

```
Connected to /74.125.67.100 on port 80 from port
2857 of /192.168.1.4
```